

Storewide Packdown Pipeline Architecture

Status

This document supersedes the overhead-to-selling hard-anchor proposal in `unified\_overhead\_selling\_pipeline.pdf`.

The correction follows the July 16 meeting clarification: an inventory box may be stored directly overhead, in another bay, or elsewhere in the store. Physical proximity between an overhead box and a selling shelf is therefore not product correspondence.

Problem statement

The systems answer different questions:

- The selling-area system determines which SKU is missing and where it must be stocked.
- The single-line and multiline OCR systems determine which SKU-bearing boxes were observed and where those boxes were seen.

The integration objective is not to use overhead OCR to infer the selling grid. It is to locate inventory for a SKU that the selling workflow has already identified as missing.

Correct data flow

```
```mermaid
flowchart LR
 A[Selling-area processor] --> B[Missing SKU request]
 B --> C[SKU + target selling location]

 D[Single-line box OCR] --> E[Storewide inventory index]
 F[Multiline box OCR] --> E
 G[Other box detections] --> E

 C --> H[Exact-SKU packdown lookup]
 E --> H
 H --> I[Inventory candidates]
 I --> J[Downstream availability and routing]
 J --> K[Packdown task]
```
```

Contracts

Missing SKU request

Produced from the selling-area empty/missing result:

```
```json
{
 "sku": "1000000456",
 "store_number": "0244",
 "selling_aisle": "08",
 "selling_bay": "003",
 "selling_location_id": 12,
 "selling_shelf_index": 1,
 "selling_position_index": 4,
 "confidence": 0.91
}
```

### ### Inventory observation

Produced by single-line or multiline box OCR:

```
```json
{
  "sku": "1000000456",
  "store_number": "0244",
  "inventory_aisle": "09",
  "inventory_bay": "017",
  "inventory_bbox": [90, 10, 150, 50],
  "photo_location_path": "gs://inventory/box.jpg",
  "photo_timestamp": "1770339128729",
  "process_source": "singleline",
  "confidence": 0.98
},
,
```

The selling location and inventory location are intentionally separate.

Matching rules

1. Match only within the same store.
2. Use an exact normalized SKU for automatic inventory lookup.
3. Search all inventory observations storewide; do not restrict by selling bay.
4. Return all matching box observations as candidates.
5. Do not automatically convert a candidate into a stocking task. Inventory availability, duplicate-box resolution, routing, and employee assignment are downstream responsibilities.
6. Three-digit suffixes and fuzzy OCR reads may support search or review, but they must not create automatic packdown tasks without an additional disambiguation policy.

Explicit non-goals

- No overhead-bbox to selling-price-label mapping.
- No nearest-bbox or vertical-column correspondence.
- No overhead OCR hard anchors in the selling matcher.
- No assumption that overhead stock mirrors the assortment below it.
- No cross-store inventory match.

Implementation in this workspace

- `output/sku\_payload.py` publishes rich `inventory\_observations` while retaining the existing `skus\_list` and `bounding\_boxes` fields.
- `pipeline/packdown\_matching.py` adapts OCR payloads, builds a storewide exact-SKU inventory index, converts selling empty-grid output into missing-SKU requests, and returns inventory candidates.
- `selling\_processor.py` remains price/sequence based and does not consume overhead OCR results.

Remaining production boundaries

The production selling handler and the downstream packdown task service are not present in this workspace. Runtime integration still requires:

1. The selling owner to publish the missing-SKU request.
2. Single-line and multiline consumers to publish inventory observations into a storewide index.
3. A packdown service to perform the exact-SKU lookup and apply live inventory,

routing, and task-assignment rules.

Validation plan

Measure:

- Missing-SKU request precision from the selling system.
- Exact-SKU OCR precision for inventory observations.
- Storewide inventory lookup recall.
- Percentage of requests with one, multiple, or zero candidate boxes.
- Candidate-location accuracy.
- Packdown completion rate and time-to-restock.
- False task rate; the target for automatic exact-SKU tasks should be near zero.